

AAYUSH KUMAR

Software Engineer | Full-Stack, AI & Enterprise Systems

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PROFESSIONAL SUMMARY

Software engineer with 9+ years across IBM, Rocket Software, and DXC. I currently work on IMS modernization and performance at IBM, and before that I helped build watsonx Code Assistant for Z, IBM's generative-AI product for mainframe modernization. Outside of work I build full-stack products with AI, from React Native and TypeScript on the front end to Python and Node.js services on Supabase. I know how large banks and insurers run, and I know how to bring modern AI into that kind of environment without breaking it.

CORE SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, COBOL, REXX, Assembler

Backend & APIs: REST API design, Python services, Node.js, Supabase (PostgreSQL, Auth, Edge Functions), API integration, webhooks

AI & MCP: Model Context Protocol (MCP) server development, Anthropic Claude API, LLM agent design, prompt engineering, AI-assisted developer tooling

Frontend & Mobile: React, React Native (Expo), TypeScript, responsive UI, dashboard design

Cloud & DevOps: Vercel, Turborepo monorepos, Cloudflare DNS, GitHub, CI/CD, Ansible, ZOAU automation

Enterprise & Performance: z/OS, IMS DB/DC, DB2, VSAM, OpenTelemetry, performance benchmarking & certification, RMF, production incident triage

PROFESSIONAL EXPERIENCE

IMS System Programmer, Modernization & Performance

2024 – Present

IBM | Bangalore

- Work with one of Spain's largest banks to resolve high-severity IMS performance problems and guide their modernization across observability and workload optimization.
- Designed and built a z/OS MCP Server and an IMS Logs MCP Server that let LLM agents work with mainframe operations and log data. Both were recognized as innovation projects in my performance review.
- Led the GA-critical OpenTelemetry validation for IMS, running baseline versus instrumented benchmarks to confirm trace enablement stayed within performance tolerances at production scale.
- Owned performance validation for IBM Z Digital Integration Hub (ZDIH), covering workload analysis and early risk identification across release stages.
- Wrote a Python and REXX framework to validate JCL transformation outputs for functional equivalence, and rebuilt our regression automation.
- Built a performance dashboard the team relies on, which cut reporting and analysis time by more than half.
- Wrote a published IBM Community article on Java workload optimization on z/OS, documenting JVM, WLM, and Db2 tuning that reduced general-purpose CPU consumption by roughly 65% in testing ([read it](#)).

AI/ML Engineer, watsonx Code Assistant for Z

2023 – 2024

IBM | Bangalore · [JCL explanation \(IBM docs\)](#)

- Worked on watsonx Code Assistant for Z (WCA4Z), IBM's generative-AI product for mainframe modernization, focused on JCL explanation and transformation.
- Defined the JCL explanation and transformation logic the model needed to produce, turning mainframe domain knowledge into clear requirements.
- Generated and curated JCL training data so the model handled a wider range of real-world job patterns.
- Built and ran evaluations to measure the quality of the model's output and fed the results back into each iteration.
- Did well enough on WCA4Z to be moved onto the IMS modernization and performance team.

Mainframe Infrastructure Engineer

Apr 2022 – Jul 2024

Rocket Software | Bangalore · [Rocket Data Virtualization](#)

- Worked directly with the customer to install Rocket Data Virtualization Manager on their z/OS system end to end, from planning and configuration through to a working deployment.
- Set up secure hybrid data access from IMS, DB2, and VSAM so distributed applications could reach mainframe data over SQL and REST.
- Automated configuration and operational tasks with Python, REXX, and Ansible, and handled SMP/E maintenance and change cycles in production.

DXC Technology | Indore

- Built and maintained COBOL and JCL modules in Life/70 policy administration systems for Allstate Benefits (voluntary and supplemental insurance).
- Supported production batch cycles, fixing failures and performance problems, and automated recurring batch operations and job monitoring to cut manual work.

SELECTED PROJECTS

QuantumShield — An AI agent that plans post-quantum cryptography migrations. Includes a learning animation that walks you through how the migration works, so the concept is easy to follow. CBOMkit, NIST PQC, React.

z/OS & IMS Logs MCP Servers — MCP servers that connect LLM agents to mainframe operations and log analysis. Python, MCP SDK.

Artha, Mosaic, Settl, Dhruva — Personal finance and money-splitting apps built with AI on React Native, TypeScript, and Supabase.

CERTIFICATIONS

IMS Fundamentals · IMS Diagnostics · z/OS Mainframe Practitioner · Assembler Programming · IBM Z Digital Integration Hub · Ansible Automation · SMP/E Foundation 2.5 · LOMA 281 (Insurance Domain)

EDUCATION

Bachelor of Engineering, Computer Science — Institute of Engineering & Technology, Indore (DAVV)	2017
Diploma, Computer Science & Engineering — Samrat Ashok Technological Institute, Vidisha (RGPV)	2014